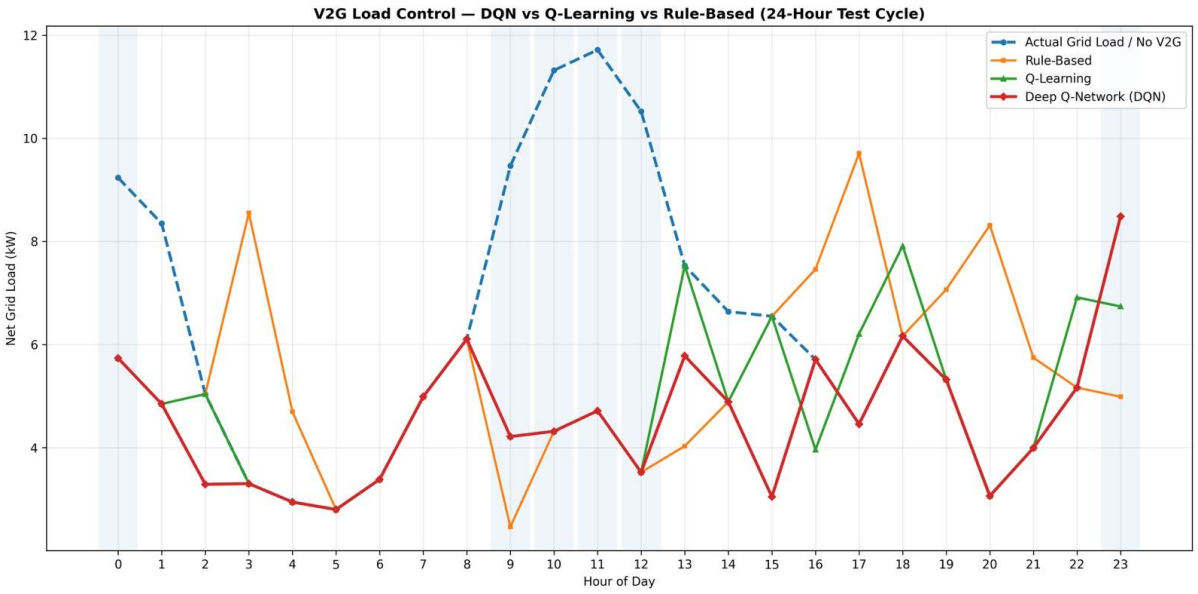
Comparison-Graph and Plot




```
import numpy as np
import matplotlib.pyplot as plt

# 24-hour simulation
hours = np.arange(24)

# 1. Actual Grid Load (Without V2G)
actual_load = [
    9.2, 8.35, 5.05, np.nan, np.nan, np.nan,
    np.nan, np.nan, 6.1, 9.45, 11.3, 11.7,
    10.5, 7.55, 6.65, 6.55, 5.7,
    np.nan, np.nan, np.nan, np.nan, np.nan,
    np.nan, np.nan
]

# 2. Rule-Based Control
rule_based = [
```

```
    np.nan, np.nan, 5.05, 8.55, 4.7, 2.8,
    3.4, 5.0, 6.1, 2.5, 4.3, 4.7,
    3.5, 4.0, 4.9, 6.55, 7.45, 9.7,
    6.15, 7.05, 8.3, 5.75, 5.2, 5.0
]

# 3. Q-Learning Control
q_learning = [
    np.nan, 4.85, 5.05, 3.3, np.nan, np.nan,
    np.nan, np.nan, np.nan, np.nan, np.nan,
    np.nan, np.nan, 7.55, 4.9, 6.55, 3.95,
    6.2, 7.9, 5.3, np.nan, np.nan, 6.9, 6.75
]

# 4. Deep Q-Network (DQN)
dqn = [
    5.75, 4.85, 3.3, 3.3, 2.95, 2.8,
    3.4, 5.0, 6.1, 4.2, 4.3, 4.7,
    3.5, 5.8, 4.9, 3.05, 5.7, 4.45,
    6.15, 5.3, 3.05, 4.0, 5.2, 8.5
]

# Create graph
plt.figure(figsize=(18, 8))

# Highlight operating periods
shaded_regions = [
    (0, 0.45),
```

```
    (9, 9.45),
    (10, 10.45),
    (11, 11.45),
    (12, 12.45),
    (23, 23.45)
]

for start, end in shaded_regions:
    plt.axvspan(
        start - 0.45, end,
        color='lightblue',
        alpha=0.20
    )

# Plot actual grid load
plt.plot(
    hours, actual_load,
    color='tab:blue',
    linestyle='--',
    marker='o',
    linewidth=3,
    label='Actual Grid Load / No V2G'
)

# Plot Rule-Based
plt.plot(
    hours, rule_based,
    color='tab:orange',
```

```
    marker='s',
    linewidth=2.2,
    label='Rule-Based'
)

# Plot Q-Learning
plt.plot(
    hours, q_learning,
    color='tab:green',
    marker='^',
    linewidth=2.5,
    label='Q-Learning'
)

# Plot DQN
plt.plot(
    hours, dqn,
    color='red',
    marker='D',
    linewidth=3,
    label='Deep Q-Network (DQN)',
    zorder=5
)

# Title
plt.title(
    'V2G Load Control — DQN vs Q-Learning vs Rule-Based (24-Hour Test Cycle)',
    fontsize=16,
```

```
    fontweight='bold'
)

# Labels
plt.xlabel('Hour of Day', fontsize=12)
plt.ylabel('Net Grid Load (kW)', fontsize=12)

# Axis limits
plt.xticks(hours)
plt.xlim(-1.5, 24.5)
plt.ylim(2, 12.2)

# Grid and legend
plt.grid(True, alpha=0.25)
plt.legend(loc='upper right', fontsize=11)

plt.tight_layout()

# Display
plt.show()
```